ECON 770 Recitation 4

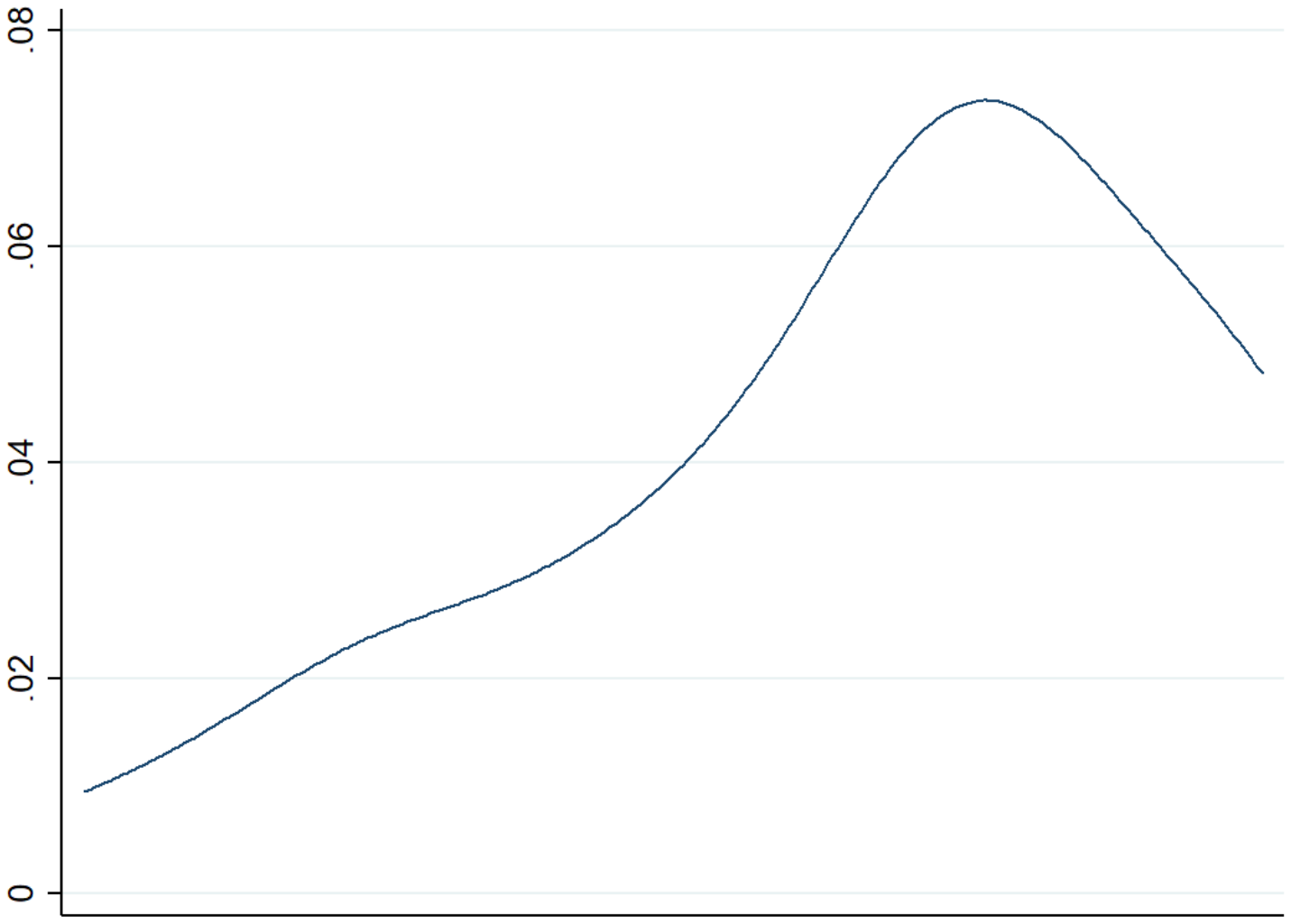
Sept 4, 2020

University of North Carolina at Chapel Hill

Reminder

- Due of Assignment #2: Sep 8, 2020 12:00 PM

Illustration



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Example of Transformation

Suppose that $X \sim U[-1, 1]$ and $Y = g(X) = X^2$, then how to solve the density function of Y?

Example of Transformation (Contd)

Suppose that $X \sim U[-1,1]$ and $Y = g(X) = X^2$, then how to solve the density function of Y?

Theorem

$$f_Y(y) = \sum_{i=1}^{k} f_Y^{(i)}(y) = \sum_{i=1}^{k} \mathbb{1}_{\{y \in S_{Yi}\}} f_X\left(g_i^{-1}(y)\right) \left|\frac{d}{dy} g_i^{-1}(y)\right|$$

Let $A_0 = \{0\}$, $A_1 = [-1,0)$ and $A_2 = (0,1]$, then on A_1 and A_2, $g(x) = x^2$ is monotonic and we can find inverse functions for $g_1(x)$ and $g_2(x)$.

$$g_i^{-1}(y) = \begin{cases} -\sqrt{y} & \text{for } i = 1 \quad (y \in S_{Y1} = (0,1]) \\ +\sqrt{y} & \text{for } i = 2 \quad (y \in S_{Y2} = (0,1]) \end{cases}$$

Then, $|\frac{d}{dy} g_i^{-1}(y)| = \frac{1}{2\sqrt{y}}$, and by theorem

$$f_Y(y) = \frac{1}{2}\left|-\frac{1}{2\sqrt{y}}\right| \mathbb{1}_{\{0<y\leq 1\}} + \frac{1}{2}\left|\frac{1}{2\sqrt{y}}\right| \mathbb{1}_{\{0<y\leq 1\}} = \frac{1}{2\sqrt{y}} \mathbb{1}_{\{0<y\leq 1\}}$$

Expectation (Definition)

Definition

- (Discrete) Suppose that $X \sim P(x)$ where $P(x) = P(X = x)$ is a p.m.f., then the expected value of X (if exists) is defined as

$$\mathbb{E}[X] = \sum_{x \in S_x} xP(X = x)$$

- (Continuous) Suppose that $X \sim f(x)$ where $f(x)$ is a p.d.f., then the expected value of X (if exists) is defined as

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} xf(x)dx = \int_{S_x} xf(x)dx$$

Expectation (Discrete Example)

Suppose we have a fair six-sided die and its outcome is denoted as X. Then $P(X = x) = \frac{1}{6}$ for $x \in \{1, 2, 3, 4, 5, 6\}$. Recall that $\mathbb{E}[X] = \sum_{x \in S_x} xP(X = x) = \sum_{x=1}^{6} xP(X = x)$, then

$$\mathbb{E}[X] = 1 \cdot \frac{1}{6} + 2 \cdot \frac{1}{6} + 3 \cdot \frac{1}{6} + 4 \cdot \frac{1}{6} + 5 \cdot \frac{1}{6} + 6 \cdot \frac{1}{6} = 3.5$$

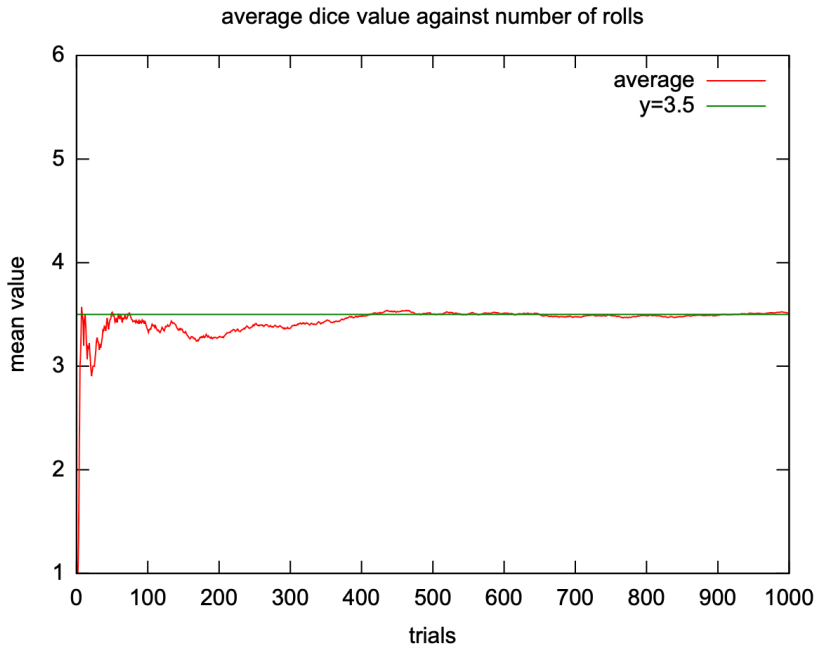


Figure: An illustration of the convergence of sequence averages (Wikipedia)

Expected Values of Various Distributions

Distribution	Notation	Mean $\mathbb{E}[X]$	Variance $Var[X]$
Bernoulli	$X \sim b(1,p)$	p	$p(1-p)$
Binomial	$X \sim B(n,p)$	np	$np(1-p)$
Poisson	$X \sim \text{Poisson}(\lambda)$	λ	λ
Uniform	$X \sim U(a,b)$	$(a+b)/2$	$\frac{1}{12}(b-a)^2$
Exponential[1 2]	$X \sim \exp(\lambda)$	$1/\lambda$ or λ	$1/\lambda^2$ or λ^2
Normal	$X \sim N(\mu, \sigma^2)$	μ	σ^2
Std. Normal	$X \sim N(0,1)$	0	1
Log-normal	$\ln(X) \sim N(\mu, \sigma^2)$	$\exp\left(\mu + \frac{\sigma^2}{2}\right)$	...
Chi-squared	$X \sim X^2(k)$	k	$2k$

[1] If $f(x) = \mathbb{1}\{x \geq 0\}\frac{1}{\lambda}e^{-x/\lambda}$, then $\mathbb{E}[X] = \lambda$ (in lecture)

[2] If $f(x) = \mathbb{1}\{x \geq 0\}\lambda e^{-\lambda x}$, then $\mathbb{E}[X] = 1/\lambda$

Properties of Expectation

Theorem

Suppose that R.V. $X \sim F_X,\ Y \sim F_Y$ with finite expectation, then

- $\mathbb{E}[aX + bY + c] = a\mathbb{E}[X] + b\mathbb{E}[Y] + c$ for any $a, b, c \in \mathbb{R}$
- If $X \leq Y$ almost surely[a], then $\mathbb{E}[X] \leq \mathbb{E}[Y]$
- If X and Y are **independent**[b], then $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$
- If $Y = g(X)$ where $g(\cdot) : \mathbb{R} \longrightarrow \mathbb{R}$ is a measurable[c] function, then

$$\mathbb{E}[Y] = \mathbb{E}[g(X)] = \int_{S_x} g(x) f_X(x) dx \text{ or } \sum_{x \in S_x} g(x) P(X = x)$$

$$= \int_{S_y} y f_Y(y) dy \text{ or } \sum_{y \in S_y} y P(Y = y)$$

[a] i.e. $P(X \leq Y) = 1$. It's also true if we let $X(\omega_1) \leq Y(\omega_2)$ for all ω_1 and ω_2

[b] Inversely we can only get $cov(X, Y) = 0$

[c] "Good" functions

Moments

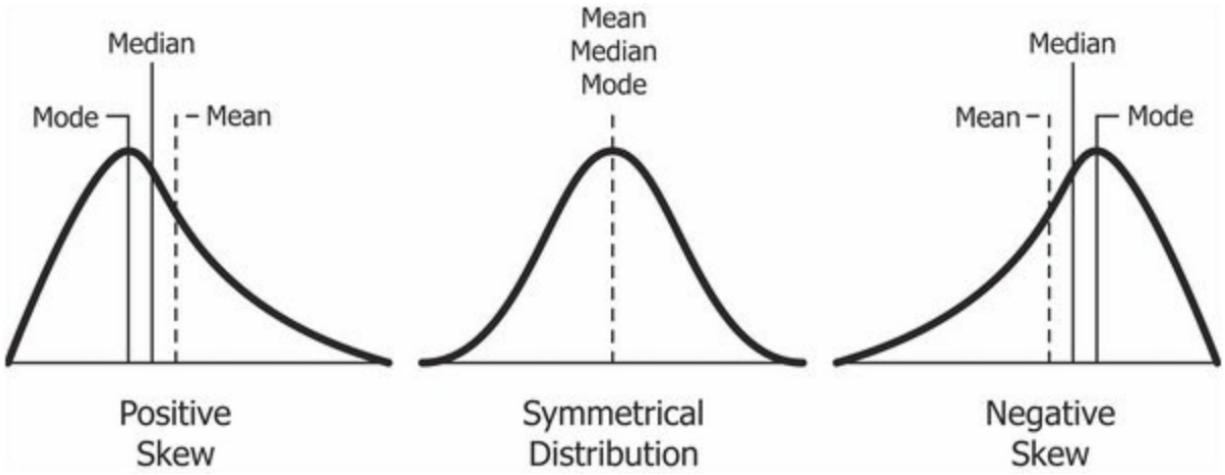
Definition

Suppose that R.V. $X \sim F$ with mean value μ, then

- m-th moment is defined as $\mathbb{E}[X^m]$
- m-th central moment is defined as $\mathbb{E}[(X-\mu)^m]$
- If X is a R.V. of Degenerate distribution, then $P(X = c) = 1$ for a constant c

Moments (Contd)

- 1st Moment: <u>Mean</u> $\mu = \mathbb{E}[X]$ weighted by density importance as $xf(x)$ or $xP(X = x)$
- 2nd Central Moment: <u>Variance</u> $\sigma^2 = \mathbb{E}[(X-\mu)^2] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$ average "distance" to mean
- 3rd Standardized Moment: <u>Skewness</u> $\tilde{\mu}_3 = \mathbb{E}\left[\left(\frac{X-\mu}{\sigma}\right)^3\right]$ measure of the asymmetry



- 4th Standardized Moment: <u>Kurtosis</u> $\text{Kurt}[X] = \mathbb{E}\left[\left(\frac{X-\mu}{\sigma}\right)^4\right]$ measure of the "tailedness"

Moments (Contd)

- Degenerate distribution has 0 variance (non-stochastic?)
- For continuous R.V. $X \sim F$, <u>median</u> m is defined s.t.

$$0.5 = P(X \leq m) = \int_{-\infty}^{m} f(x)dx$$

- *Example of mean and median*
  Suppose that $X \sim \exp(\lambda) : f(x) = \frac{1}{\lambda}e^{-x/\lambda}\mathbb{1}\{x \geq 0\}$, then

$$\mathbb{E}[X] = \frac{1}{\lambda}\int_{0}^{\infty} xe^{-x/\lambda}dx \text{ and } m = \lambda \ln(2) \text{ by } 0.5 = \int_{0}^{m} \frac{1}{\lambda}e^{-x/\lambda}dx$$

- *Example of heavy tail*
  Suppose that $X \sim$ *Pareto*, then $f(x) = 4x^{-5}$ and $\mathbb{E}[X^4] = \infty$

Symmetry

Definition

$X \sim f(x)$ is <u>symmetric at c</u> if for any $\varepsilon \in \mathbb{R}$, $f(c+\varepsilon) = f(c-\varepsilon)$ (or $P(X = c+\varepsilon)$

Theorem

If X has a symmetric distribution at c then $c = \mu = m$ where μ is the mean and m is the median. Moreover, for odd r, $\mathbb{E}[(X-c)^r] = 0$ (thus $\tilde{\mu}_3 = 0$).

How to prove this?

- In general, let $x = c + z$, split integral and plug in $f(c+z) = f(c-z)$
- If we know $f(x)$ (e.g. $U[0,1]$), we can compute μ, c and m to verify the symmetry.

Inequality

Theorem

- Chebyshev's Inequality

$$P(|X| > c) \leq \frac{1}{c^2}\mathbb{E}|X|^2$$

- Markov's Inequality

$$P(|X| > c) \leq \frac{1}{c^k}\mathbb{E}|X|^k$$

- Jensen's Inequality
  - $g(\mathbb{E}[X]) \leq \mathbb{E}[g(X)]$ if $g(\cdot)$ is convex
  - $g(\mathbb{E}[X]) \geq \mathbb{E}[g(X)]$ if $g(\cdot)$ is concave
- Absolute Value $|\mathbb{E}[X]| \leq \mathbb{E}|X|$
- Liapunov's Inequality $(\mathbb{E}[X]^p)^{1/p} \leq (\mathbb{E}[X]^r)^{1/r}$ $(p \leq r)$

Moment Generating Function

Definition

The Moment Generating Function of R.V. $X \sim F$ is defined as

$$m(\lambda) = \mathbb{E}\left[e^{\lambda X}\right]$$

X is bounded if $\exists\, M > 0$ s.t. $P(|X| < M) = 1$

Theorem

$$\left.\frac{\partial^r}{\partial \lambda^r} m(\lambda)\right|_{\lambda=0} = \mathbb{E}\left[X^r\right] \quad \text{if } \mathbb{E}[X^r] \text{ exists}$$

Exercise (Upenn 2019 prelim): Compute $\mathbb{E}[X^3]$ and $\mathbb{E}[X^4]$ where $X \sim N(\mu, \sigma^2)$